Dependence-Estimation Novelty Survey

Date: 2026-10-27 **Status:** Retrieved-evidence survey for the proposed Option B research framing. Not owner-approved. This document tests a candidate claim; it does not authorize implementation or alter AD-001 through AD-005.

Method: Primary-source retrieval with verbatim quotation. Every quotation below was verified against a locally retained copy of its source using `.agents/skills/research/scripts/validate_quote.py`. Source recency gate: created or updated on or after 2024-10-27. Quotations from PDF sources had line-break hyphenation, whitespace, and typographic ligatures normalized; no wording was altered.

Claim under test

> Online estimation of inter-source dependence **from claim streams alone** - no cross-covariance, no shared state channel, no ground truth - where the measured consequence is **task-lifecycle correctness**, not estimation error.

Three constraints are stacked. Each is tested separately below, because the claim survives only if no inspected source satisfies all three together.

Counterexample matrix

C1. Reliable Truth Discovery for Dynamic and Dependent Sources (IEEE TKDE 38(1), January 2026)

Establishes: HMM-based estimation of dependence among sources reporting conflicting values, including dependence that changes over time.

> "One of the key challenges centers on measuring the dynamic dependence among sources."

> "These models are based on the Hidden Markov Model (HMM) and are designed to handle different types of dependencies"

> "real-world scenarios are often more complex, with dynamic dependence relationships among sources over time"

Limitation for the candidate claim: Scored on truth-discovery accuracy against known object values. The inspected portions describe no action layer, no verification cost, and no task lifecycle. Not embodied and not visual.

Disposition: DIRECT COUNTEREXAMPLE to constraints 1 and 2. Online dependence estimation from categorical claim streams, using an HMM, is occupied.

C2. Revisiting Split Covariance Intersection (arXiv:2501.07915, January 2025)

Establishes: Conservative fusion when correlation is unknown, plus an extension that exploits known correlated components.

> "In distributed or cooperative systems, the cross-covariance terms cannot be computed, and to avoid underestimating the estimation error, conservative fusions must be performed."

> "This paper extends SCI to also take advantage of the correlated components."

Limitation: Operates on estimators carrying covariance representations. Bounds correlation rather than estimating it, and scores on covariance tightness, not on discrete decisions.

Disposition: NARROW. Dependence-aware *fusion* is occupied and actively being optimized. The candidate must not claim it.

C3. Decentralized Multi-Robot Obstacle Detection and Tracking in a Maritime Scenario (arXiv:2602.12012v2, 2026)

Establishes: Multi-robot visual detection with conservative fusion and uncertainty-driven tasking.

> "Robots exchange compact track summaries that are fused conservatively using Covariance Intersection, preserving estimator consistency under unknown cross-correlations."

> "An information-driven allocator assigns targets and selects UAV hover viewpoints by trading expected uncertainty reduction in travel effort and safety separation."

It also has a task and entity lifecycle. An earlier draft of this survey claimed it did not. That was wrong on the retained text, which contains "EKF-based track management" with explicit track pruning, an assignment state machine where "NA indicates no assignment", and an anti-churn term: the allocator includes a stickiness reward "to reduce assignment chattering" (the reward's notation is omitted here because the retrieved HTML renders its mathematics unreliably). It is evaluated on discrete lifecycle-correctness metrics: "our approach achieved perfect identity continuity with low

fragmentation and good accuracy." Fragmentation and identity continuity are structurally the same family of measure as false closure and incorrect reopening.

The one limitation that survives: it *bounds* correlation conservatively via Covariance Intersection rather than *estimating* it, and therefore never regresses a dependence estimate against its lifecycle outcomes.

Disposition: CLOSEST SYSTEM-LEVEL COUNTEREXAMPLE. It combines visual multi-robot evidence, unknown-correlation fusion, expected-uncertainty-reduction tasking, an assign/reassign lifecycle, and lifecycle-correctness scoring. PD-003 cannot be claimed as new. This source constrains the candidate claim far more tightly than the rest of the matrix.

C4. KC-Bench (arXiv:2609.03588, September 2026)

Establishes: A benchmark scoring whether agents reconcile conflicting information before acting.

> "they must reconcile user instructions, parametric knowledge, and dynamic environmental observations before taking actions"

Limitation:

> "KC-Bench isolates this model-level behavior rather than ranking complete agent frameworks"

Text and tool domains, not multi-agent visual evidence. No dependence or correlation axis.

Disposition: NARROW. Scoring conflict handling by downstream action consequences is occupied in the LLM-agent setting.

C5. General Collaborative Intelligence (arXiv:2609.22967, 2026)

Establishes: Motivating evidence that correlated failure is a real multi-agent perception problem.

> "homogeneous agents with identical perception models and similar viewpoints are susceptible to correlated failures, where all agents miss the same occluded entity; collaboration does not help when the collective shares a common blind spot."

It also supplies the strongest available statement that the evaluation gap this project targets is real:

> "this phenomenon is documented in early work on Byzantine-resilient fusion" ... "but is rarely evaluated as a benchmark metric."

(The elision is a citation marker. The phenomenon referred to is a fused collective world model becoming worse than what each agent would achieve independently.)

It is not only a survey. An earlier draft of this survey said it supplied no evaluation. That was wrong: "we propose GCI-Bench, a five-pillar scoring protocol with a maturity model that makes the trade-offs of collaborative methods comparable across studies."

It also describes claim-stream-only reliability estimation and independence-conditioned gating. Trust-aware fusion frameworks "address this by assigning time-varying reliability scores to each node based on latency, spatial consistency, and historical performance" — no ground truth, no cross-covariance. And it describes ROBOSAC [96], "accepting only detections confirmed by multiple independent viewpoints to filter out spoofed data."

Disposition: LEAD-BEARING COUNTEREXAMPLE, not motivation. Per-source reliability is not inter-source dependence, and accept/reject is not a five-state task lifecycle — but independence-conditioned discrete action gating in a visual multi-agent setting is materially closer to constraint 3 than this survey first conceded. ROBOSAC (2023) and refs [106]-[108] must be inspected before the constraint-3 verdict is relied on.

Verdict by constraint

1. **Dependence estimation from categorical claim streams** - **OCCUPIED.** C1 does this with HMMs, including dependence that changes over time. **Correction:** an earlier draft called this "online" estimation and labelled it a source claim. The word "online" appears twice in the retained C1 text, neither describing C1's own method; one refers to a *different* cited work. C1 extracts dependence via the Viterbi algorithm, which decodes a complete observation sequence. Whether C1 operates in a streaming regime was **not established** from the inspected text. "Dynamic" is the source's claim; "online" was this survey's inference.
2. **Without cross-covariance or ground truth** - **OCCUPIED.** The covariance-intersection line (C2, C3) avoids needing cross-covariance by presuming partial structural knowledge of it; C1 estimates dependence without ground truth; C5 describes time-varying per-node reliability from the stream alone.
3. **Scored on task-lifecycle correctness and verification cost** - **NOT FOUND OCCUPIED in combination with 1 and 2, on bounded and incomplete evidence.** C4 scores action consequences but the inspected abstract shows no dependence axis. C3 has the visual setting, the lifecycle, *and* lifecycle-correctness scoring, and is excluded on one ground only: it bounds correlation rather than estimating it, so it never relates a dependence estimate to those outcomes. C5 describes independence-conditioned gating and is an open threat.

Inference (derived, not asserted by any source): on the inspected evidence, the most defensible candidate contribution appears to be neither the estimator nor the fusion rule, but the measured link

between an *estimated* inter-source dependence and *discrete task-lifecycle errors* - false closure, incorrect reopening, and verification trips - in a multi-agent visual setting.

Falsifier. A single retrieved source that regresses an estimated inter-source dependence against discrete task or track lifecycle outcomes in a multi-agent perception setting would close this gap. ROBOSAC (2023) and the closed-loop collaborative-perception evaluation line are the most likely candidates and have **not** been inspected.

Consequences for the project

- PD-001 (HMM resolver) must not be presented as novel. C1 uses HMMs for exactly this family of problem.
- PD-003 (value-of-information gate) must not be presented as novel. C3 already allocates by expected uncertainty reduction.
- AD-004 remains the right constraint, but dependence-aware fusion is an occupied solution space. The project should adopt an existing dependence treatment and measure its task-level consequence, rather than propose a new fusion rule.

Explicit non-claims

- Not a new fusion operator, and not an improvement on covariance intersection.
- Not a new dependence estimator; an existing formulation should be adopted.
- Not a claim that correlated evidence degrades task decisions; that is the hypothesis to be measured, not an established result.
- Not a field-first claim. This is bounded retrieval over the sources listed below.

Limitations of this survey

This survey has not completed its adversarial-review cycle. It was reviewed once, failed, and was corrected on the points above. The constraint-3 verdict has not yet been re-tested against the leads the review surfaced.

- **Three of five sources were read partially.** C1: first six pages of thirteen; its scoring on truth-discovery accuracy is supported by its abstract, not by its experiments section. **C4: the arXiv abstract page only (~6 KB)** — the claim that it has no dependence axis is made from ~200 words, and its abstract does list "multi-source temporal conflicts", a category that could involve correlated sources. C5: mined selectively from ~1 MB.

- **The recency gate is a search budget, not a novelty boundary.** It distorts in both directions. It excludes the foundational covariance-intersection and data-incest literature. It also hides that constraint 1 is not a 2026 result: C1's own abstract notes that "Existing studies primarily focus on dependent sources with conflicting information, where sources may copy information from each other" — a lineage running back to the VLDB 2009 copy-detection work. Most damagingly, ROBOSAC (2023) sits inside C5, is pre-gate, and bears directly on constraint 3.
- Searches were English-language against general web indexes, not a systematic database protocol. This is bounded search evidence, not proof of absence.

Open actions before this survey is citable

1. Inspect ROBOSAC (ICCV 2023) and C5 refs [106]-[108].
2. Run an ungated search on constraint 3 specifically: closed-loop collaborative-perception evaluation (V2X-Sim and successors), and V2X misbehaviour detection, checking whether any models *inter-source* dependence rather than per-source reliability.
3. Retrieve the KC-Bench full paper before any claim about its dependence axis.
4. Resolve C1's reference [20], described there as "A novel online truth discovery framework", which bears directly on the online-versus-dynamic correction above.

Sources

1. Zhang, H., Wang, S., Chen, L., Li, X., Gao, Q., Sheng, Q. Z. "Reliable Truth Discovery for Dynamic and Dependent Sources." *IEEE Transactions on Knowledge and Data Engineering* 38(1), January 2026. http://web.science.mq.edu.au/~qsheng/papers/TKDE-2026-truth-discovery.pdf
2. "Revisiting Split Covariance Intersection: Correlated Components and Optimality." arXiv:2501.07915, January 2025. https://arxiv.org/abs/2501.07915
3. "Decentralized Multi-Robot Obstacle Detection and Tracking in a Maritime Scenario." arXiv:2602.12012v2, 2026. https://arxiv.org/html/2602.12012v2
4. Lyu, Y., Zhou, S., Toh, B., Zhu, P., Li, L. "KC-Bench: A Dynamic Interactive Benchmark for Evaluating Knowledge Conflicts in LLM Agents." arXiv:2609.03588, September 2026. https://arxiv.org/abs/2609.03588
5. "General Collaborative Intelligence: Architecting Cognition for Resilient Multi-Agent Ecosystems." arXiv:2609.22967, 2026. https://arxiv.org/abs/2609.22967

Consulted but not quoted: DIFNet (arXiv:2508.18854, August 2025); Overlapping Covariance Intersection (arXiv:2603.16768v2, 2026).